

Comparative study of WCSPH, EISPH and explicit incompressible-compressible SPH (EICSPH) for multi-phase flow with high density difference

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ABSTRACT

This study presents three Smoothed Particle Hydrodynamics (SPH) methods capable of handling high-density differences in violent incompressible multiphase flows. The conventional Weakly Compressible SPH (WCSPH) is reformulated into a quasi-Lagrangian framework based on Arbitrary Lagrangian-Eulerian (ALE) context. The Explicit Incompressible SPH (EISPH) method is extended to handle multiphase flows and reformulated in the ALE framework. The Explicit Incompressible-Compressible SPH (EICSPH) method, which can handle both compressible and incompressible phases, is developed by combining these two approaches in a fully explicit algorithm. The proposed methods are validated and compared through four benchmark problems including Rayleigh-Taylor instability, hydrostatic, liquid sloshing and dam break problems. Firstly, the stability and accuracy of interface modeling of the three methods were verified through Rayleigh-Taylor instability with small density differences. In the hydrostatic problem with a large density difference, the results from all three methods exhibit good agreement in the pressure distribution. Particularly, EICSPH demonstrates the faster convergence when compared to WCSPH, with EISPH showing fastest convergence overall. In the transient sloshing problem, all three methods quantitatively converged to the experimental results and exhibited good agreement, although slight pressure noise was observed in WCSPH initially. Finally, in the dam break problem, all three methods successfully simulated sharp interfaces without non-physical voids. The temporal variations of pressure and height were well predicted by all three methods, with EISPH showing better conformity to the water-air interface morphology obtained from other incompressible numerical methods compared to WCSPH and EICSPH. Additionally, the impact of air speed of sound was examined in EICSPH, where while the difference in pressure variation at the probe was minimal, differences were observed in energy dissipation and the progression of the free surface due to the air cushion effect.

1. Introduction

The simulation of multi-phase flows, which are ubiquitous in both scientific and engineering domains, holds significant importance in numerous applications in various fields [1–4]. One of the biggest challenges in multi-phase flow simulation is to accurately capture the interface between the different phases, especially when there is a high density difference between the fluids. The grid-based

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numerical methods such as finite volume (FV), finite difference (FD) and finite element (FE) typically require special algorithms such as the volume of fluid (VOF), level set (LS) and front tracking to tract the interface between different phases. The smoothed particle hydrodynamics (SPH) method is a deterministic meshless Lagrangian approach that has emerged as a promising alternative for simulating incompressible multi-phase flows. Its strength in tracking interfaces without the need for additional algorithms has led to its widespread use in various fields [5–9].

Early SPH researches focused on simulating complex interfacial flows in single-phase fluids. The most general approach is the weakly compressible SPH (WCSPH), which allow a small compressibility. Two methods are used to compute density variation in WCSPH [10]: the summation approach and the continuity approach. However, it is well-known that the continuity approach suffers from high-frequency noise. In order to eliminate the pressure noise, several methods are adopted including periodic re-initialization [11], diffusion terms [12] and Riemann solvers [13,14]. In recent years, there has been increasing interest in multi-phase flow simulation using SPH and several studies has been carried out to address the stability issues and pressure noise induced from large density discontinuity. To mitigate the density discontinuity in the summation approach, the density summation is reformulated based on number density [15] or normalized form with reference density [16]. For the continuity approach, modifications have been made to the continuity equation through reformulation with discrete gradient based on volume [11,17,18], reformulation of diffusion term [16,19,20], and the incorporation of a multi-phase Riemann solver [21,22]. However, some past researches explored some instability in simulating multiphase flows due to non-uniform particle distribution at phase interfaces or non-physical void formation [23,24]. Several approaches have been proposed to address these issues. Background pressure is adopted to avoid pairing instability from negative pressure [18,25] and a small amount of artificial repulsive force is applied at the phase interface to prevent particles of different phases penetrating each other [18,22]. In addition, particle shifting technique (PST) is introduced in multi-phase flows. The PST is widely recognized and employed for enhancing accuracy and stability in particle methods in terms of single-phase free surface and internal flows [26–29]. Recently, PST is extended to multi-phase WCSPH method to keep uniform particle distribution [23], and it has been further extended to the projection-based particle method [30].

The other possible choice to simulate fluid flow is Incompressible SPH (ISPH) [31]. ISPH is known for its noise-free pressure field and the ability to handle large time-steps compared to WCSPH [32]. Typically, ISPH is solved with semi-implicit algorithm [31], but there have been limited researches that have investigated multi-phase flows with high density difference (density ratios up to 1000) based on ISPH algorithm [33–35]. The computation of Pressure Poisson Equation (PPE) in ISPH is known to be both computationally intensive and challenging to parallelize [36]. As an alternative approach, an explicit ISPH (EISPH) method has been proposed [37–39]. EISPH has been shown to have lower pressure noise compared to WCSPH and low computational cost compared to ISPH [40,41], and has been extended to multi-phase flows with high density difference in very recent studies [42,43]. Another alternative to simulate multi-phase flow with ISPH is the two-phase compressible-incompressible method, where the gaseous and liquid phases are treated as compressible and incompressible, respectively [44–47]. This approach has been utilized to simulate multi-phase flows with high density differences and investigate the effects of the gaseous phase [45,46] and pressure fluctuations in compressible bubbles submerged in incompressible flows [44,47].

This paper proposes three improved methods for simulating multi-phase flows with high density differences. The first method reformulates the conventional multi-phase WCSPH approach [16] as a quasi-Lagrangian scheme based on simplified ALE formulation to improve stability. The second method is an extension of the efficient EISPH approach for multi-phase flows with large density differences, and it is also formulated as a quasi-Lagrangian scheme using ALE formulation. The third method is the Explicit-Incompressible-Compressible SPH (EICSPH), which couples both explicit multi-phase SPH methods and treats the gaseous phase as compressible and the liquid phase as incompressible. To evaluate the computational efficiency and stability of these three methods, we compare them against four benchmark problems including Rayleigh-Taylor instability, hydrostatic, liquid sloshing and dam break problem.

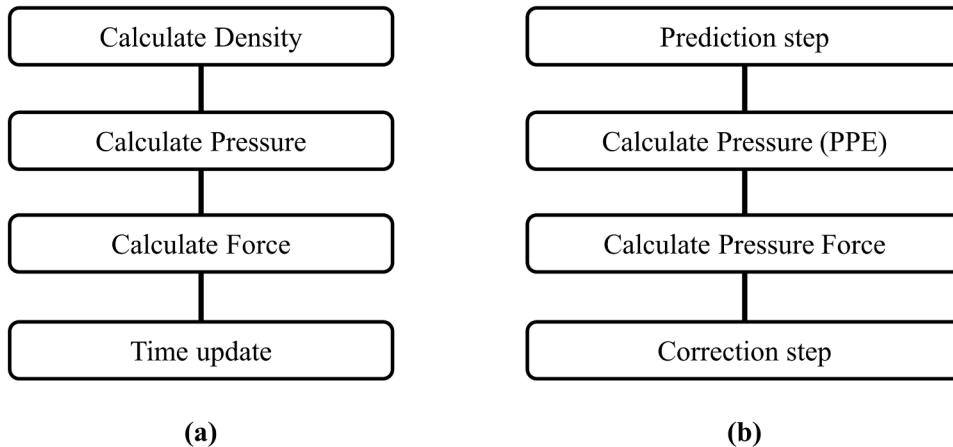


Fig. 1. Flowcharts of (a) weakly compressible approach and (b) incompressible approach.

The remaining sections of this paper are structured as follows. Section 2 provides an overview of two approaches to solving incompressible flow using the governing equations. In Section 3, we present the methodology for SPH discretization and a comprehensive derivation of the proposed multi-phase models. Section 4 presents the results of benchmark problems to demonstrate the accuracy and performance of our approach. Finally, in Section 5, we summarize our conclusions and suggest directions for future work. In this study, we utilized an open-source multi-physics SPH code called SOPHIA [16,48], which is written in CUDA C language and fully parallelized by graphics processing unit (GPU).

2. Governing equations for incompressible flows

This section provides a concise overview of the governing equations for two approaches to solve incompressible flow: weakly compressible assumption and incompressible assumption. The overall algorithms for each approach are summarized in Fig. 1. The discretized forms of the governing equations using SPH approximation will be introduced in Section 3.

2.1. Weakly compressible assumption approach

The governing equations for the motion of an incompressible, isothermal, and Newtonian fluid in the Lagrangian framework comprise of mass conservation and momentum conservation equations, expressed as follows:

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{u}, \quad (1)$$

$$\frac{D\mathbf{u}}{Dt} = -\frac{\nabla P}{\rho} + \nu \nabla^2 \mathbf{u} + \mathbf{g}, \quad (2)$$

where, Dt/D represents the Lagrangian derivative, ρ and $\mathbf{u}$ are physical variables denoting density and velocity, respectively. P represents pressure, μ represents viscosity, and $\mathbf{g}$ denotes gravitational acceleration. The system of governing equations can be closed by explicit computation of the equation of state (EOS) under the assumption of weak compressibility of density. The Tait EOS is employed in this study [10,49] and described as:

$$P = \frac{\rho_{ref} c_0^2}{\gamma} \left(\left(\frac{\rho}{\rho_{ref}} \right)^\gamma - 1 \right) + P_b, \quad (3)$$

where γ is the polytropic index of phase, P_b is a background pressure, ρ_{ref} and c_0 are reference density and speed of sound. In air, γ is typically around 1.4, while for water it is approximately 7. The numerical speed of sound of water phase is adjusted to allow for a compression ratio of approximately within 3 %. The numerical speed of sound in the air phase is assigned with real speed of sound and it is further discussed in subsequent sections.

2.2. Incompressible assumption approach

In the incompressible approach, Eq. (1) leads to

$$\nabla \cdot \mathbf{u} = 0. \quad (4)$$

The first order projection method [31,50] derived based on the Helmholtz-Leroy decomposition [51] is commonly used to compute pressure field. In the projection method, the forces are solved in two steps. First, the auxiliary velocity is predicted based on forces excluding the pressure force as Eq. (5). Secondly, the corrected velocity is updated by pressure force as, Eq. (6).

- The prediction step

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t (\nu \nabla^2 \mathbf{u} + \mathbf{g}) \quad (5)$$

- The correction step

$$\mathbf{u}^{n+1} = \mathbf{u}^* + \Delta t \left(-\frac{\nabla P^{n+1}}{\rho} \right) \quad (6)$$

The superscript $*$ denote the auxiliary step and the superscript n denote number of the time step. Between the two steps, it is necessary to compute the pressure field that enforce the incompressibility constraint. The Eq. (6) can be rewritten as:

$$\nabla P^{n+1} = -\rho \frac{\mathbf{u}^{n+1} - \mathbf{u}^*}{\Delta t}. \quad (7)$$

The velocity field in the subsequent step must satisfy the divergence-free condition ($\nabla \cdot \mathbf{u}^{n+1} = 0$), and upon substitution into Eqs. (7), and (8) can be derived. Furthermore, Eq. (8) can be transformed into Eq. (9) by first-order time integration of density ($\rho \nabla \cdot \mathbf{u}^* = (\rho - \rho^*)/\Delta t$). Here, ρ^* is intermediate density based on SPH approximation in Eq. (11) and the density ρ remains constant, equal to the reference density, as ISPH does not allow compressibility. The Eq. (8) is commonly referred to as the divergence-free (DF) condition, while Eq. (9) is termed the density-invariance (DI) condition [36].

$$\nabla^2 \mathbf{p}^{n+1} = \rho \frac{\nabla \cdot \mathbf{u}^*}{\Delta t} \quad (8)$$

$$\nabla^2 \mathbf{p}^{n+1} = \frac{\rho - \rho^*}{\Delta t^2} \quad (9)$$

However, in this study, the governing equations are reconstructed based on the ALE formulation from the next section and therefore only DF condition is used to compute the PPE.

3. Numerical method

In this section, we provide an overview of the discretization method of smoothed particle hydrodynamics (SPH) and present the derivation and discretization of three distinct multiphase SPH schemes.

3.1. SPH approximation

The basic SPH approximation of a scalar function $f(\mathbf{r})$ at a point $\mathbf{r}$ is defined as the convolution over a support domain Ω by using a weighting function $W(\mathbf{r} - \mathbf{r}', h)$ with a smoothing length h .

$$f(\mathbf{r}) = \int_{\Omega} f(\mathbf{r}') W(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' \quad (10)$$

The continuous fluid system is represented by discretized particles, and the function $f(\mathbf{r})$ of the i^{th} particle is written as

$$f_i := f(\mathbf{r}_i) = \sum_j f_j W_{ij}(|\mathbf{r}_{ij}|, h) V_j = \sum_j f_j W_{ij}(|\mathbf{r}_{ij}|, h) \frac{m_j}{\rho_j}, \quad (11)$$

where subscript j denotes the neighboring particles, m , ρ and V are the mass, density and volume, respectively. Hereinafter, the weighting function $W_{ij}(|\mathbf{r}_{ij}|, h)$ with $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ will be simplified as W_{ij} . In this study, Wendland C6 kernel function is adopted as the weighting function owing to its outstanding stability [52,53].

$$\begin{cases} W(q) = C(1-q)^8(1+8q+25q^2+32q^3) & \text{if } 0 \leq q < 1 \\ W(q) = 0 & \text{if } q > 1 \end{cases} \quad (12)$$

Here, $q = r/(\kappa h)$ and $C = 78/(7\pi(\kappa h)^2)$ in 2D. κ is the scale factor which defines the effective area of the kernel smoothing function and it is set as 3 in our study. The smoothing length $h = 1.5\Delta x$, where Δx is the initial particle spacing. Therefore, the particle of interest (i^{th} particle) interacts with neighboring particles within a support radius of κh . The general form of SPH gradient, divergence and Laplacian operators of a scalar function f_i and vector function $\mathbf{F}_i$ are derived as follows [54]:

$$\nabla f_i = \sum_j [f_j \pm f_i] \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (13)$$

$$\nabla \cdot \mathbf{F}_i = \sum_j [\mathbf{F}_j \pm \mathbf{F}_i] \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (14)$$

$$\nabla^2 f_i = \sum_j 2[f_j - f_i] \frac{\mathbf{r}_{ij} \cdot \nabla_i W_{ij}}{|\mathbf{r}_{ij}|^2} \frac{m_j}{\rho_j}. \quad (15)$$

3.2. Multi-phase WCSPH

3.2.1. Lagrangian description

The SPH approximation of governing equations for fluid flow in Eqs. (1) and (2) are derived in past studies [10,55]. The multiphase WCSPH model derived based on normalized density [16] is adopted and written as:

$$\frac{D\rho_i}{Dt} = -\rho_i \sum_j \mathbf{u}_{ji} \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta h c_0 D_i^{\rho,nd}, \quad (16)$$

$$\frac{D\mathbf{u}_i}{Dt} = \mathbf{g} - \sum_j 2\bar{P}_{ij}\nabla_i W_{ij} \frac{m_j}{\rho_i \rho_j} + \mu_i \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (17)$$

where $\mathbf{u}_{ji}$ denotes the differences $\mathbf{u}_j - \mathbf{u}_i$ and the inter-particle pressure $\bar{P}_{ij} := (P_i + P_j)/2$. The diffusion term $D_i^{\rho,nd}$ is represented as

$$D_i^{\rho,nd} := \sum_j 2\psi_{ij}^{nd} \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (18)$$

$$\psi_{ji}^{nd} := \rho_{ref,i} \left\{ \left(\frac{\rho_j}{\rho_{j,ref}} - \frac{\rho_i}{\rho_{i,ref}} \right) - \frac{1}{2} \left(\left\langle \nabla \frac{\rho_i}{\rho_{ref,i}} \right\rangle^L + \left\langle \nabla \frac{\rho_j}{\rho_{ref,j}} \right\rangle^L \right) \cdot (\mathbf{r}_{ji}) \right\} \frac{\mathbf{r}_{ji}}{|\mathbf{r}_{ji}|^2}, \quad (19)$$

where the superscript ‘L’ denotes that the gradient of the density is reformulated based on the renormalized matrix $\mathbf{L}_i$, which is expressed as [56]:

$$\left\langle \nabla \frac{\rho_i}{\rho_{ref,i}} \right\rangle^L = \sum_j \left(\frac{\rho_j}{\rho_{j,ref}} - \frac{\rho_i}{\rho_{i,ref}} \right) \mathbf{L}_i \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad \mathbf{L}_i := \left[\sum_j \mathbf{r}_{ji} \otimes \nabla_i W_{ij} \frac{m_j}{\rho_j} \right]^{-1}. \quad (20)$$

The parameter δ is set to be 0.1 for all simulations [12]. The viscous force term Π_{ij} is represented as [57].

$$\Pi_{ij} = 2(d+2) \frac{\mathbf{u}_{ij} \cdot \mathbf{r}_{ij}}{|\mathbf{r}_{ij}|^2 + \eta^2}, \quad (21)$$

where d denotes the spatial dimension and η is a small constant set to be $0.1h$ to avoid singularity problem.

3.2.2. ALE description

The above approach is here modified to quasi-Lagrangian formulation based on ALE time derivative and it is given by [46,58–62]:

$$\frac{df}{dt} := \frac{\partial f}{\partial t} + (\mathbf{u} + \delta \mathbf{u}) \cdot \nabla f = \frac{Df}{Dt} + \delta \mathbf{u} \cdot \nabla f. \quad (22)$$

Here, $\partial t / \partial$ represents the Lagrangian derivative and $\delta \mathbf{u}$ is the particle shifting velocity. When $\delta \mathbf{u}$ is zero, the framework is purely Lagrangian. From the previous work on ALE formulation based on WCSPH (Yoo et al., 2022), the continuity equation can be written as:

$$\frac{d\rho_i}{dt} = -\rho_i \sum_j \mathbf{u}_{ji} \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta \mathbf{u}_i \cdot \sum_j \rho_{ji} \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta h c_0 D_i^{\rho,nd}, \quad (23)$$

where ρ_{ji} denotes $\rho_j - \rho_i$. However, the convection term added as the second term in the above equation becomes unphysically large when computing multiphase flow with large density difference due to the discontinuity of density. Therefore, the convection term is reformulated in terms of the normalized density as in the diffusion term and the set of governing equations (Eqs. (16) and (17)) are

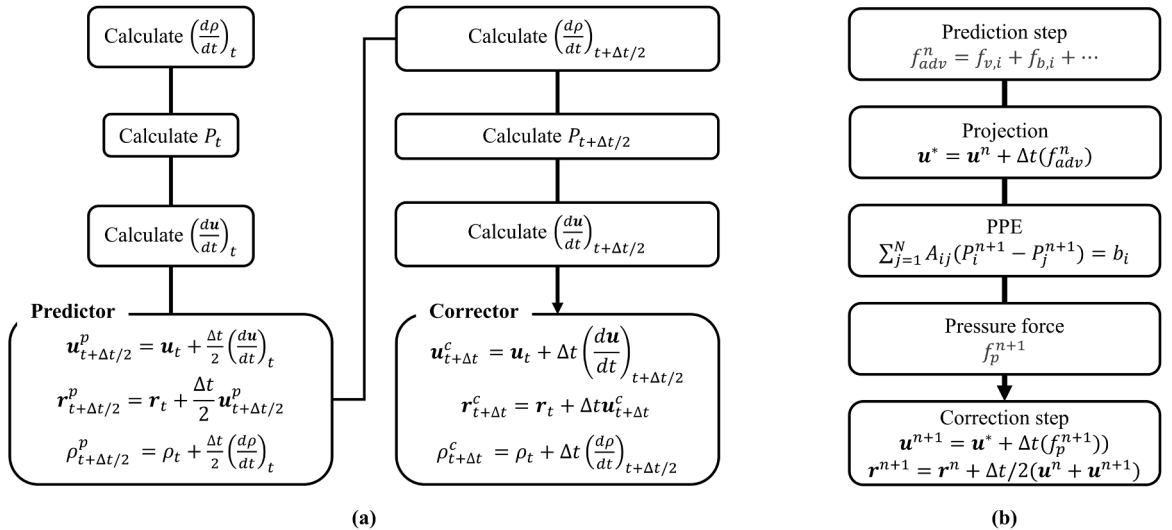


Fig. 2. Overall algorithms for (a) multi-phase WCSPH scheme and (b) multi-phase EISPH scheme.

rewritten as:

$$\frac{d\rho_i}{dt} = -\rho_i \sum_j \mathbf{u}_{ji} \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta \mathbf{u}_i \rho_{ref,i} \sum_j \left(\frac{\rho_j}{\rho_{j,ref}} - \frac{\rho_i}{\rho_{i,ref}} \right) \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta h c_0 D_i^{\rho,nd}, \quad (24)$$

$$\frac{d\mathbf{u}_i}{dt} = \mathbf{g} - \sum_j 2\bar{P}_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_i \rho_j} + \mu_i \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta \mathbf{u}_i \cdot \sum_j \mathbf{u}_{ji} \otimes \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (25)$$

$$m_i = \rho_i V_{0,i} \left(\frac{V_i}{V_{0,i}} \right) = \rho_i V_{0,i} \left(\sum_j \frac{m_j}{\rho_j} \frac{1}{V_{j,0}} W_{ij} \right) / \left(\sum_j W_{ij} \right), \quad (26)$$

where V_0 corresponds to the initial volume of a particle equal to the initial spacing(Δx) raised to the power of dimension(d). The governing equations are integrated in time with 2nd order predictor-corrector as in Fig. 2a [63]. The time step is bounded based on two constraints: *CFL* (Courant-Friedrichs-Levy) condition and viscous condition which are given by [64]:

$$\Delta t \leq \left(CFL_{WCSPH} \frac{h}{c_0}, 0.125 \frac{h^2}{\nu} \right), \quad (27)$$

where CFL_{WCSPH} is set as 0.3 in this study. To solve multiphase flows with high density differences, previous studies have typically required large sound speeds for lighter fluids to ensure stable results, resulting in smaller time steps compared to single-phase simulations [11,18]. Recently, novel multiphase models have been developed that allow for the use of small sound speeds for both air and water, improving computational efficiency [13,25,65]. However, in this study, these novel models were not employed and actual sound speed was used for the air phase to better evaluate the air-cushion effect as underlined by recent studies [14,66]. Further discussions on air-cushion effect are described in Section 4.

3.3. Multi-phase EISPH

3.3.1. Lagrangian description

Using the SPH Laplacian operators in Section 2, the discretized form of PPE is written as:

$$\sum_{j=1}^N \left(P_i^{n+1} - P_j^{n+1} \right) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h)}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} = \frac{\sum_j \rho_i \mathbf{u}_{ji}^* \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t}. \quad (28)$$

Applying the first step of the Jacobi method as in previous studies [37,36], an explicit formulation of the PPE in EISPH is obtained as:

$$\sum_{j=1}^N \left(P_i^{n+1} - P_j^n \right) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h)}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} = \frac{\sum_j \rho_i \mathbf{u}_{ji}^* \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t} \quad (29)$$

However, the original formulation has been observed to encounter numerical instabilities, particularly when the density difference is large [35]. In order to tackle this issue, prior studies have explored alternative formulations of the PPE and employed the Laplacian operator based on a corrective 2nd order tensor [35,42]. In this study, the above formulation is revisited as follow [43]:

$$\sum_{j=1}^N \left(P_i^{n+1} - P_j^n \right) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h)}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} = \frac{\sum_j \bar{\rho}_{ij} \mathbf{u}_{ji}^* \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t}, \quad (30)$$

where the $\bar{\rho}_{ij}$ denotes the inter-particle density between i^{th} and j^{th} particles. While previous studies on multiple phases have introduced arithmetic mean form [33,67], this study utilizes the harmonic mean form. It is widely recognized in previous literatures, that using the harmonic mean offers better performance than using the arithmetic mean when defining inter-particle properties [68,69].

$$\bar{\rho}_{ij} := \frac{2\rho_i \rho_j}{\rho_i + \rho_j} \quad (31)$$

The discrete governing equations can be summarized in the following three steps:

- The prediction step

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = \mathbf{g} + \mu_i \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j} \quad (32)$$

- PPE

$$\sum_{j=1}^N \left(P_i^{n+1} - P_j^n \right) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h)}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} = \frac{\sum_{j \neq i} \frac{2\rho_i \rho_j}{\rho_i + \rho_j} \mathbf{u}_{ji}^* \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t}, \quad (33)$$

• **The correction step**

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^*}{\Delta t} = - \sum_j 2\overline{P^{n+1}}_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_i \rho_j}. \quad (34)$$

In the multi-phase EISPH, the inter-particle pressure is evaluated as [67,70]:

$$\overline{P^{n+1}}_{ij} := \frac{P_i^{n+1} \rho_i + P_j^{n+1} \rho_j}{\rho_i + \rho_j}. \quad (35)$$

3.3.2. ALE description

The above approach is here updated to ALE description based on ALE time derivative. Following the projection method derived from the Helmholtz decomposition theorem, the intermediate velocity field requires the subtraction of the conservative term (pressure gradient term) from the velocity in the next time-step. Therefore, the convection term from ALE description is added in the prediction step to get intermediate velocity:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = \mathbf{g} + \mu_i \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j} + \delta \mathbf{u}_i \cdot \sum_j \mathbf{u}_{ji} \otimes \nabla_i W_{ij} \frac{m_j}{\rho_j}. \quad (36)$$

The governing equations are integrated in time based on an above explicit projection method as in Fig. 2b. The time step is bounded by the following condition:

$$\Delta t \leq \left(CFL_{EISPH} \min \left(\frac{h}{\|\mathbf{u}_i\|} \right), 0.125 \frac{h^2}{\nu} \right), \quad (37)$$

where CFL_{EISPH} is 0.075 in this study. Due to the larger time step and the simplicity of the algorithm, EISPH was approximately ten times faster than WCSPH for the same problem.

3.4. EICSPH

The two previously discussed methods are now combined into a single approach which is named as Explicit Incompressible Compressible SPH (EICSPH), where one phase is treated as incompressible and the other as compressible. To be consistent, the inter-

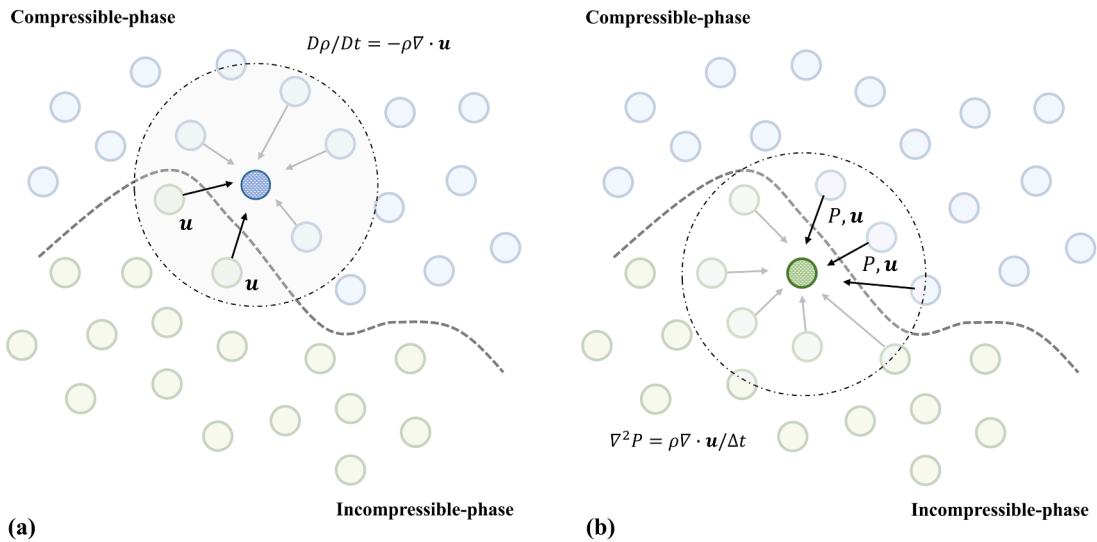


Fig. 3. Interaction between compressible phase and incompressible phase in pressure computation step of (a) Compressible particles and (b) Incompressible particles.

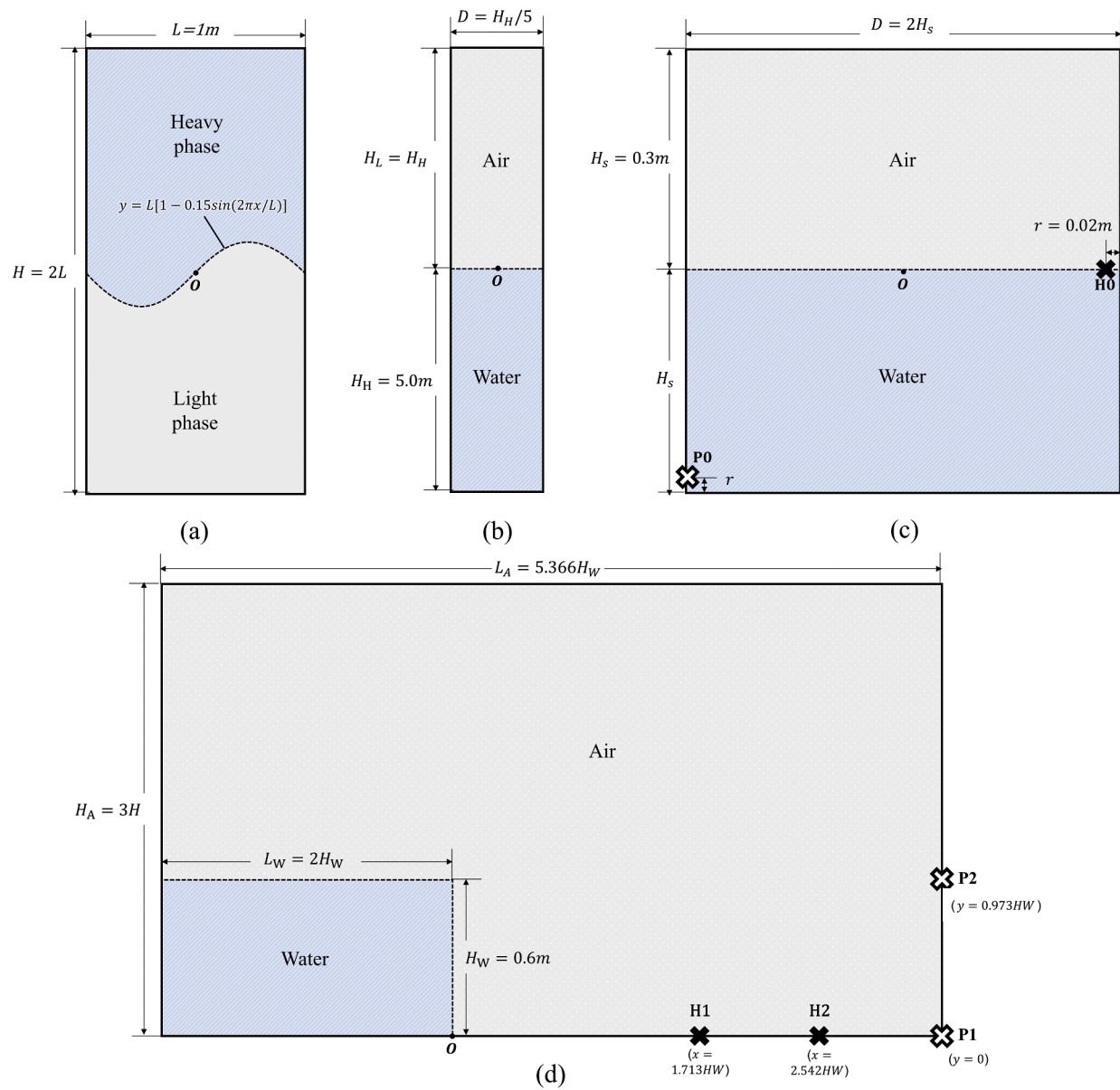


Fig. 4. Initial configuration of two benchmarks (a) Rayleigh-Taylor Instability (b) Hydrostatic (c) Sloshing (d) Dam-break.

particle pressure is evaluated by the same formulation of $\overline{P}_{ij}$ in both incompressible and compressible phase. Although each phase employs a different algorithm to compute the flow and pressure fields, it is crucial for the two phases to be mathematically coupled to ensure the continuity of physical properties and achieve realistic simulation as demonstrated by previous research [46].

In a coupled system, the particles in the compressible phase are solved, based on a predictor-corrector time scheme as in Fig. 2a. The particles in the incompressible phase are solved based on the explicit projection method as in Fig. 2b. The real sound speed is utilized in the compressible phase similar to WCSPH case, and each phase adheres to their respective time constraints as in Eqs. (27) and (37). Although incompressible particles can have larger time-step compared to WCSPH, in general, due to the higher number of compressible particles compared to incompressible particles, there is usually not a significant improvement in computation time, and the relative time difference compared to WCSPH is problem-dependent. More details are discussed in Subsection 4.5.

The pressure and velocity fields of compressible particles are used to solve the PPE and act as boundary conditions in the incompressible particles (Fig. 3a). Conversely, the velocity field of the incompressible particles is used to solve the continuity equation in the compressible particles (Fig. 3b). When solving the momentum equation for each phase, the velocity and pressure fields from the other phase are utilized.

3.5. Numerical treatment

To ensure particle uniformity, the particle shifting technique (PST) is employed as a particle shifting displacement [28,58,61]. Based on the Ficks law, the particle shifting velocity is computed using the modified concentration gradient to prevent tensile instability [28].

$$\nabla_i C_i = \sum_j \left[1 + 0.2 \left(\frac{W_{ij}}{W_{ij}(\Delta x)} \right)^4 \right] \frac{m_j}{\rho_j} \nabla_i W_{ij} \quad (38)$$

$$\delta \mathbf{u}_i = -D_i' \nabla_i C_i \quad (39)$$

Many studies have proposed similar but different diffusion coefficients D_i' . The velocity deviation in the present work is written as [61,71]:

$$\delta \mathbf{u}_i = -\kappa h_i |\mathbf{u}_i| \nabla_i C_i. \quad (40)$$

To guarantee the particle shifting velocity smaller than physical velocity, the magnitude of shifting velocity is limited to be smaller than a half of magnitude of velocity field based on following formulation.

$$\delta \mathbf{u}_i = \begin{cases} -\kappa h_i |\mathbf{u}_i| \nabla_i C_i & \text{if } |\kappa h_i |\mathbf{u}_i| \nabla_i C_i| < 0.5 |\mathbf{u}_i|, \\ -0.5 |\mathbf{u}_i| \frac{\nabla_i C_i}{|\nabla_i C_i|} & \text{otherwise,} \end{cases} \quad (41)$$

In order to ensure the kinematic boundary condition at the interface and avoid distortion of the interfacial flow particle located near the interface regions move according to the physical velocity resulting in $\delta \mathbf{u}_i = 0$. Interface particles are identified by the criterion based on the minimum eigenvalue (λ_i) of the inverse matrix of renormalization matrix (L_i^{-1}) [1–4,29,72–74]. The particles that have $\lambda_i < 0.6$ are identified as interface particles.

To avoid negative pressure and instability near the interface, background pressure and interface sharpness force are introduced. The use of background pressure is known to be very effective method to prevent particle clumping due to negative pressure and tensile instability in both single-phase [70,73,75] and multi-phase flows [18,76]. However, excessive background pressure can result in non-physical dissipation [77]. Therefore, in this study, the background pressure was kept at a level that can prevent negative pressure while being small enough compared to avoid large numerical dissipation. In order to avoid non-physical inter-penetration between different phases near the interface, Interface-sharpness force is included in the momentum equation with following formulation [17,18,76]:

$$f_{\text{interface}} = -\varepsilon \sum_j (|P_i| + |P_j|) \nabla_i W_{ij} \frac{m_j}{\rho_i \rho_j}, \quad (42)$$

where ε is a parameter in the range of 0.01–0.1.

For efficient computation, all computing steps of the three methods are implemented using single-device GPU parallelization. Further details on the parallelization strategy are elaborated in previous study [16,48]. All simulations in this study were conducted on a workstation featuring an NVIDIA Tesla A100 80GB PCIe GPU and an Intel(R) Xeon(R) Gold 6342 CPU @ 2.80 GHz.

4. Numerical benchmarks

In this study, the three proposed models are demonstrated through four benchmark test cases: Rayleigh–Taylor instability, hydrostatic, sloshing and dam-break problem. We modeled the wall boundaries as ghost particles [78] and applied the Neumann condition on pressure and slip-condition on velocity [79]. Fig. 4 depicts the initial geometric information of four benchmarks.

4.1. Rayleigh–Taylor instability

As a first benchmark, Rayleigh–Taylor instability is examined to assess the accuracy and convergence of the proposed methods under small density differences. The geometrical configuration investigated is depicted in (Fig. 4a), with the densities of the heavy and light phases set to 1800 kg/m^3 and 1000 kg/m^3 , respectively. The gravitational acceleration was set to 1 m/s^2 , and the interface condition was defined as $y = 1 - 0.15\sin(2\pi x)$. The Froude number was 1, and the Reynolds number was 420. The speed of sound for the

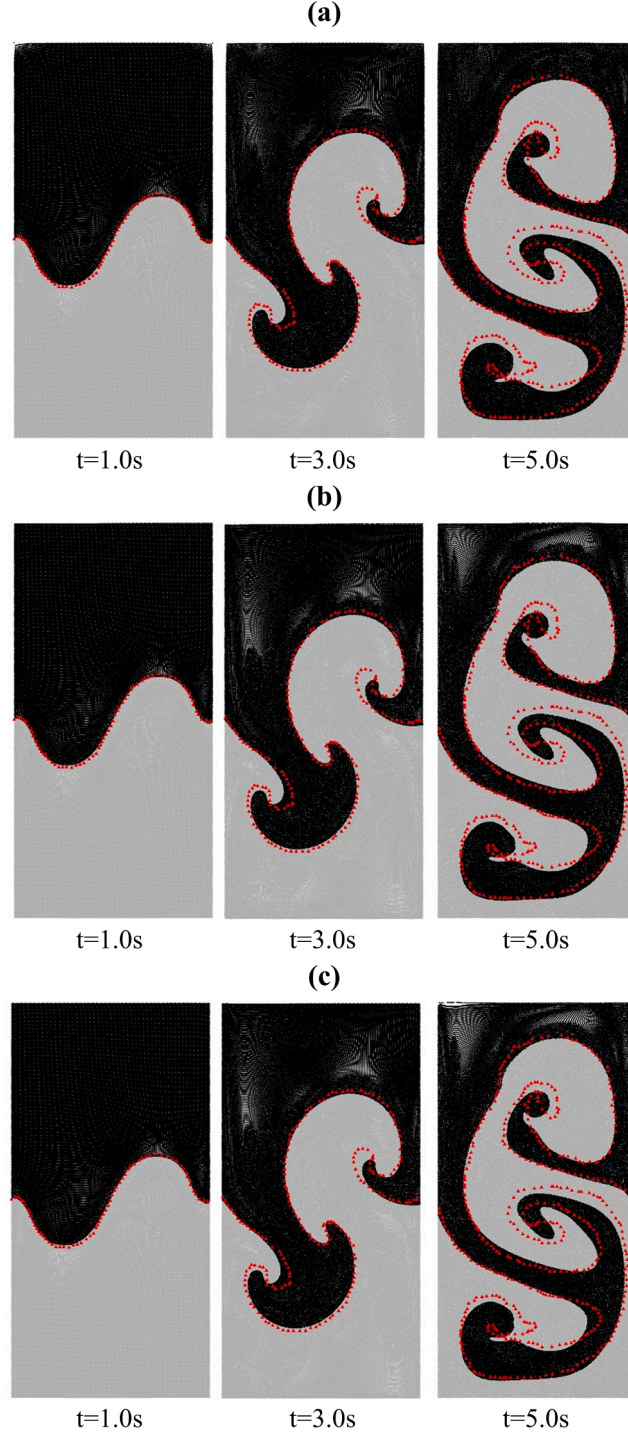


Fig. 5. Rayleigh–Taylor instability: snapshots at time instant $t = 1.0, 3.0, 5.0\text{s}$ obtained by (a) WCSPH (b) EISPH (c) EICSPH.

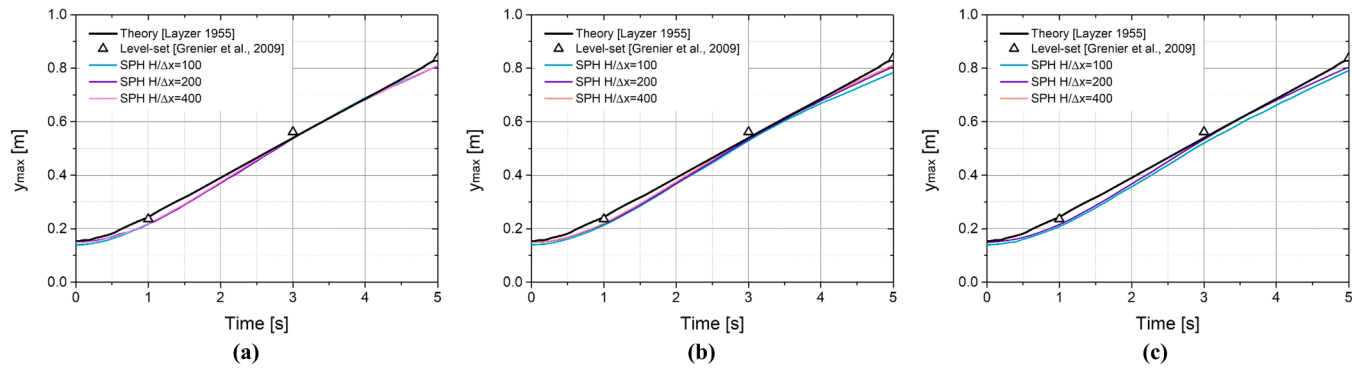


Fig. 6. Rayleigh–Taylor instability: position of maximum height of lighter phase obtained by (a) WCSPPH (b) EISPPH (c) EICSPH.

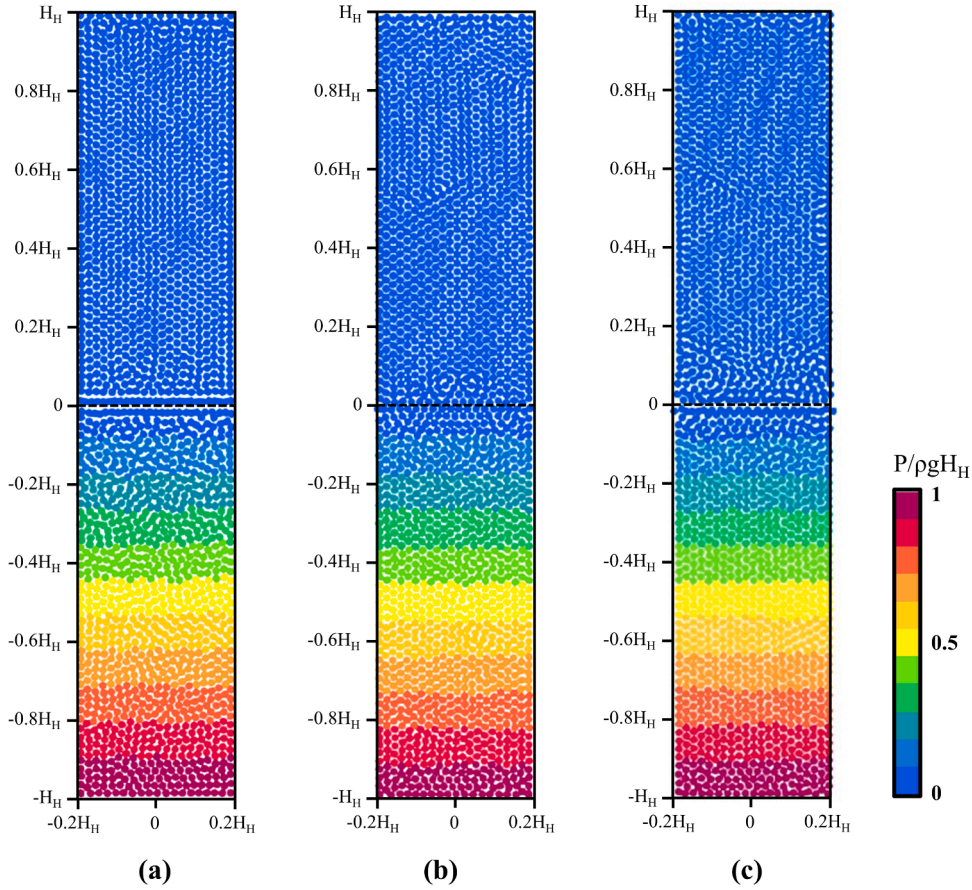


Fig. 7. Two-phase 2D hydrostatic: snapshots at time instant $t(g/H)^{0.5} = 40$ obtained by (a) WCSPH (b) EISPH (c) EICSPH.

heavy phase was 15 m/s, and for the light phase, it was set to 20 m/s. The spatial resolution is set as $\Delta x = H/50, H/100, H/200$. In Fig. 5, the intricate phase boundaries captured from the finest resolution in three methods are compared to the level set results at time instants $t = 1.0$ s, 3.0 s and 5.0 s, demonstrating similar morphologies. The maximum height of the lighter phase position results at three different resolutions were compared with theoretical predictions and the level set solution in Fig. 6. While small deviations exist between the numerical results and theoretical predictions, it is evident that as the spatial resolution increases, the results converge towards the theory.

4.2. Two-phase 2D hydrostatic

In this section, the stability and robustness of the proposed models are validated by solving the two-phase hydrostatic problem in a rectangular tank, of which the initial configuration is given in Fig. 4b. The tank is divided into two regions, with the lower half occupied by a water phase and the upper half by an air phase. The density of water and air phases are 1000 kg/m^3 and 1 kg/m^3 . Two fluids are assumed to be inviscid and particles are initially arranged on a Cartesian grid with a spatial resolution of $\Delta x/H_H = 50$. The kinetic energy of the air and water phases are plotted over time and the pressure distribution along the vertical centerline is compared to the analytical result. In this problem, the background pressure is set as $P_b = 0.1(\rho_H g H_H)$. In the case of WCSPH, the speed of sound of water phase is set as $10(g H_H)^{1/2}$. For both WCSPH and EICSPH, the actual speed of sound, which is 340 m/s, is used for the air phase.

Fig. 7 illustrates the pressure distribution of the 2D hydrostatic simulation at $t(g/H)^{0.5} = 40$. The fluid particles are shown to have a stable and uniform spatial distribution due to the effects of reference pressure and particle shifting technique. The simulation results clearly indicate a distinct water-air interface without any interpenetration, and a stable particle distribution is observed near the theoretical interface indicated by the dotted line.

In Fig. 8a, the pressure distribution is shown along the vertical centerline at $t(g/H)^{0.5} = 40$, and compared with the analytical trend. All three methods accurately capture the non-uniform pressure gradient near the water-air interface. The time history of the average pressure of the water phase obtained in the initial stage is plotted in Fig. 8b. The initial acoustic noise results in WCSPH exhibiting the slowest convergence rate, followed by EICSPH which considers compressibility of air, and the completely incompressible EISPH which shows the fastest convergence.

Fig. 9 shows the variation of kinetic energy for the light (air) and heavy (water) phases over time. In all three methods, the kinetic energy is gradually dissipated. In the water phase, it is noticed that EISPH achieves a more stable convergence than WCSPH and EICSPH. On the other hand, in the case of the air phase, EISPH exhibits a relatively high kinetic energy value at the convergence point, which is believed to be due to the alleviation of discontinuity by the change in governing equations, but not fundamentally resolved. Nevertheless, the result remains non-divergent and exhibits stable behavior throughout the long-term simulation. In contrast, for WCSPH and EICSPH, which have resolved this discontinuity, the kinetic energy gradually decays over time.

4.3. Two-phase liquid sloshing

In this section, we address the two-phase liquid sloshing problem to validate the accuracy of transient simulations with significant density disparities. The geometric configuration, depicted in Fig. 4c, follows from previous studies [21,80]. A square box of dimensions 0.6 m x 0.6 m is considered, with half filled with water and the remaining half with air. The densities of the water and air phases are 1000 kg/m³ and 1 kg/m³, respectively. In the case of WCSPH, the speed of sound of water phase is set as $10(gD)^{1/2}$, where D is the diameter of square tank. For both WCSPH and EICSPH, the actual speed of sound, which is 340 m/s, is used for the air phase. The tank undergoes horizontal periodic motion over time as below:

$$x = -A(1 - \cos \omega t), \quad (43)$$

where the amplitude A and angular velocity ω is 0.005 m and 6.85 rad/s, respectively. As illustrated in Fig. 4c, pressure probe P0 and water level measurement sensor H0 are installed and monitored, and their readings are compared with experimental results [80].

In Fig. 10, the water-air interface and pressure contours from three methods are depicted for three time instants $t(g/D)^{1/2} = 12, 24.87, 33$. Smooth pressure contours and a similar morphology of the water-air interface are observed. The pressure contours and interface shapes of EISPH and EICSPH demonstrate a high degree of similarity. However, in the case of WCSPH, small differences are shown in both the pressure distribution and interface shape qualitatively compared to other methods. The pressure signals at pressure probe P0 and height measured at height measurement sensor H0 are presented in Fig. 11 until time instant $t(g/D)^{1/2} = 30$, respectively. Upon quantitative analysis, the water surface elevation, influenced by the tank's horizontal oscillations, is accurately represented, exhibiting trends consistent with experimental observations. Furthermore, the pressure evolution consistently aligns with experimental findings across the three resolutions for all three methods. In the case of WCSPH, minor noise is observed initially, but a stable pressure field is exhibited after $t(g/D)^{1/2} = 15$. It is evident that all methods converge to pressure trends consistent with experimental results as resolution increases.

4.4. Two-phase 2D dam break

In order to validate the accuracy of the proposed models, we solved the two-phase dam-break problem. The dam-break problem is a challenging problem due to the complex free surface and vorticities generated by the impact on the wall, making it one of the most popular benchmark simulation in SPH community. The geometrical configuration is illustrated in Fig. 4d same as previous experimental studies [81,82]. The density of water and air phases are 1000 kg/m³ and 1 kg/m³. In the case of WCSPH, the speed of sound of water phase is set as $10(gH_w)^{1/2}$. For both WCSPH and EICSPH, the actual speed of sound, which is 340 m/s, is used for the air phase. The time instants of local pressure and wave heights are monitored by the pressure sensors P1, P2 at right wall, and height measurement sensors H1, H2 at bottom walls installed as in Fig. 4d. The spatial resolutions for simulation are set at $\Delta x = H_w/40, H_w/80$,

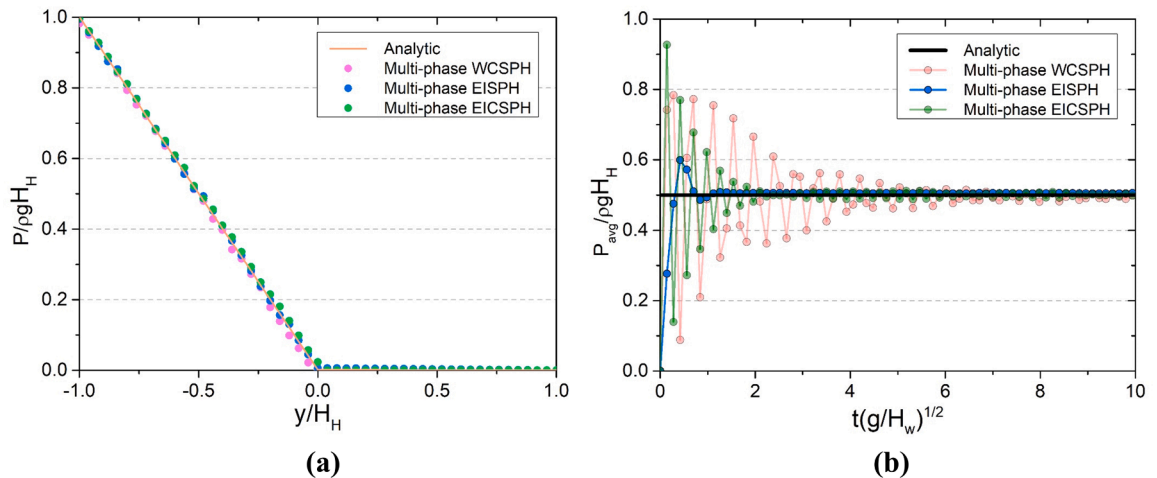


Fig. 8. Two-phase 2D hydrostatic: (a) pressure distribution along the vertical centerline at $t(g/H)^{0.5} = 40$ and (b) time history of averaged pressure of water phase at initial stage using three schemes.

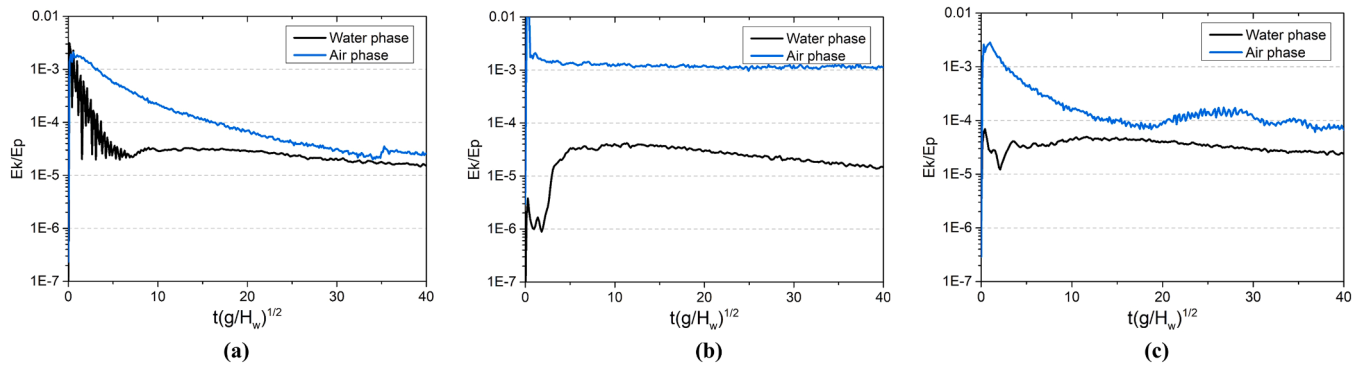


Fig. 9. Two-phase 2D hydrostatic: time evolution of kinetic energy of water/air phases obtained by (a) WCSPH, (b) EISPH and (c) EICSPH.

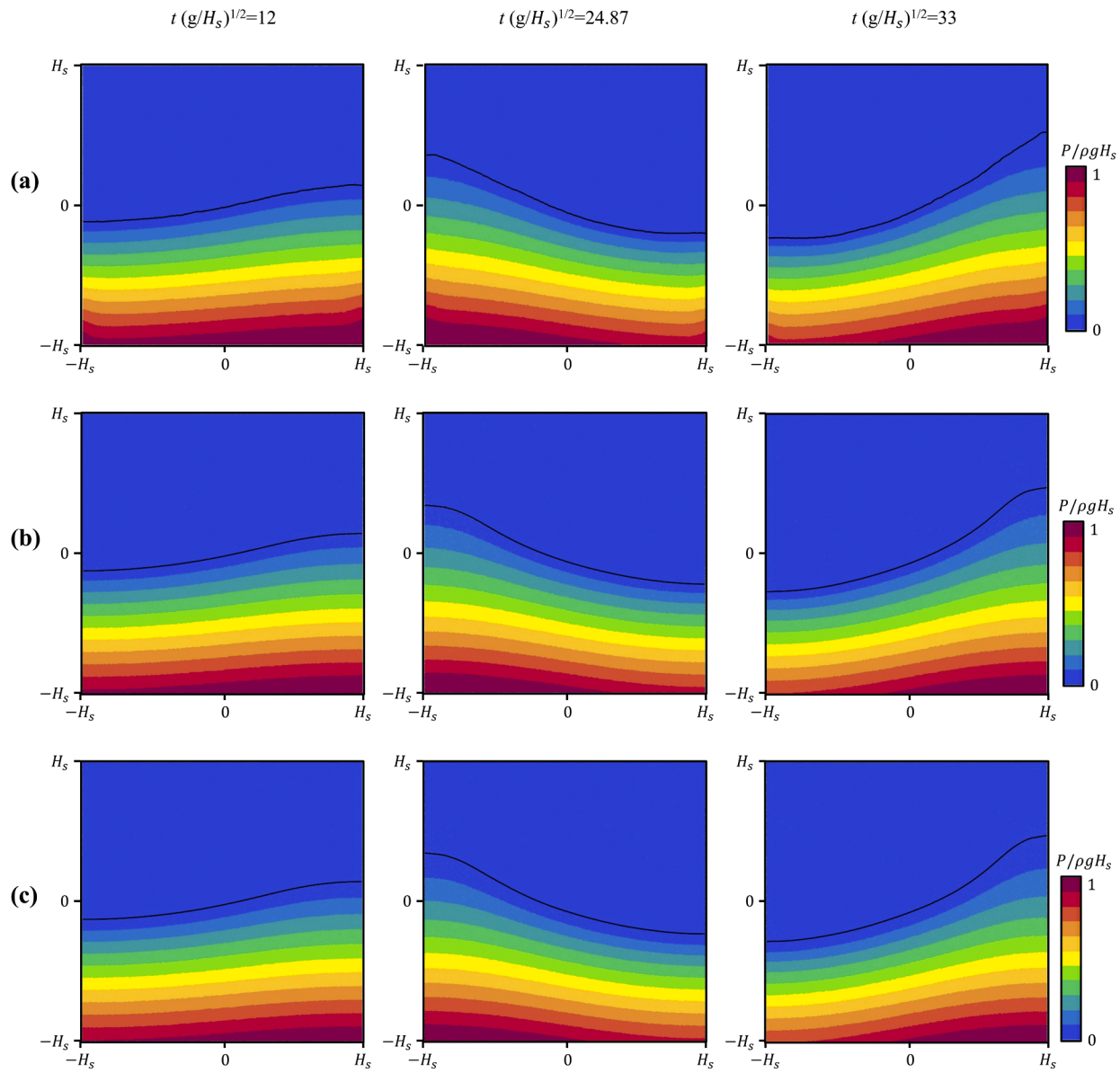


Fig. 10. Two-phase 2D liquid sloshing: pressure contours at three time instants $t(g/D)^{1/2} = 12, 24.87, 33$ by (a) WCSPH, (b) EISPH and (c) EICSPH compared with experimental results.

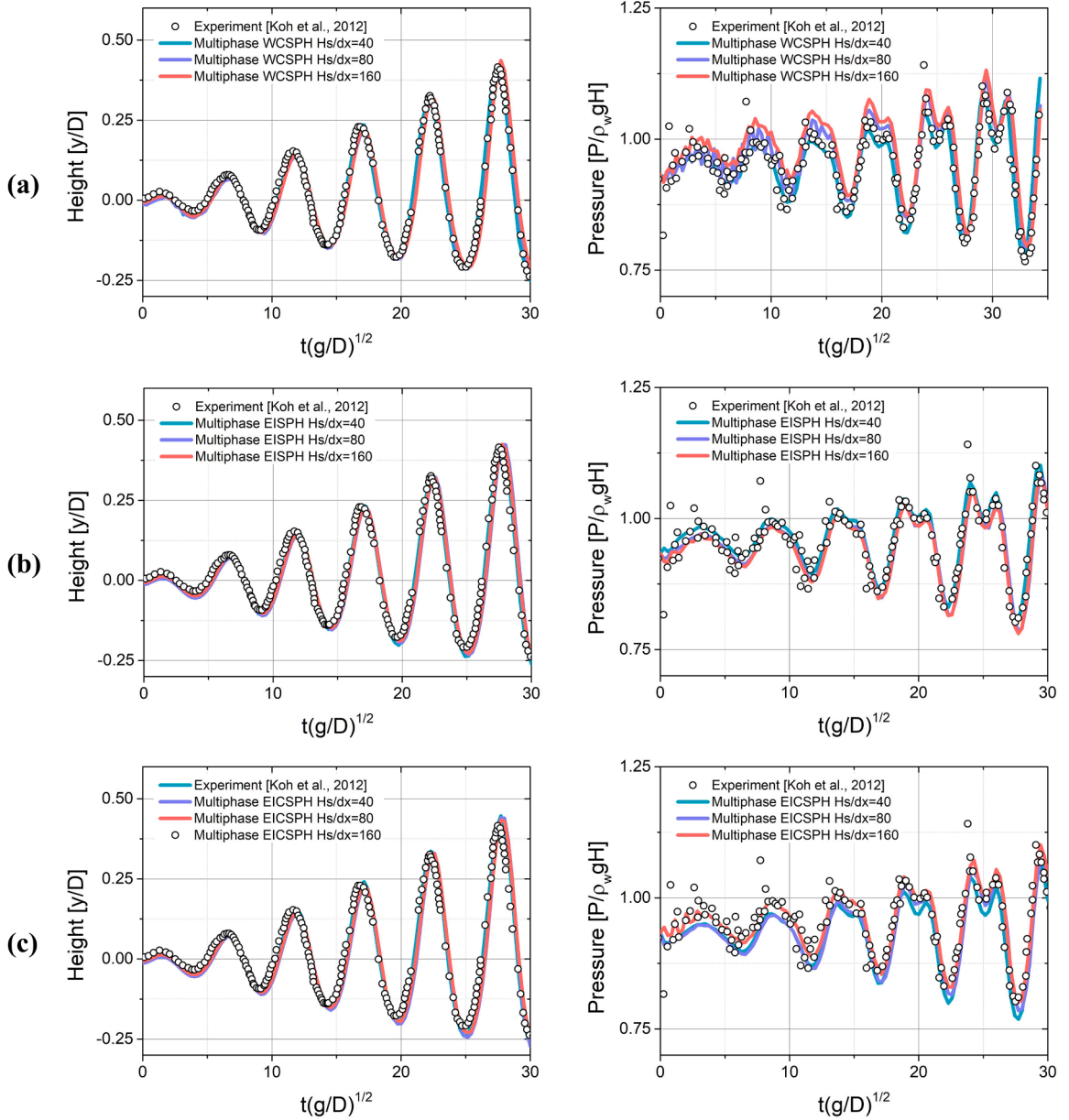


Fig. 11. Two-phase 2D liquid sloshing: the time histories of the pressure signal at pressure probe P1 and water level at height measurement sensors H0 by (a) WSPH, (b) EISPH and (c) EICSPH compared with experimental results.

$H_W/160$. Simulation results measured by each sensor are compared with experimental results over simulation time. The time instant local pressure is computed by averaging the particles within the kernel length as:

$$P_{probe} = \frac{\sum_{j \in fluid} p_j W(\mathbf{r}_{probe} - \mathbf{r}_j, h)}{\sum_{j \in fluid} W(\mathbf{r}_{probe} - \mathbf{r}_j, h)}, \quad (44)$$

where P_{probe} and $\mathbf{r}_{probe}$ are pressure and position of pressure probes, respectively. In this problem, the background pressure is set as $P_b = 0.1(\rho_H g H_W)$. In this problem, the viscosity of water and air is set as 0.05 and 0.0005 Pa·s. The initial pressure distribution is given by an analytic solution [83].

Fig. 12 shows snapshots of the computational results of WSPH, EISPH, and EICSPH over time evolution. After the dam collapses, complex flows are generated as water roll-up along the walls, splashes along the walls and collides with the water surface, creating reflected jet. Sharp water-air interfaces are well captured as in previous single-phase and multi-phase studies. Non-physical voids and penetration between phases which are reported in some previous studies are effectively prevented by the use of reference pressure and

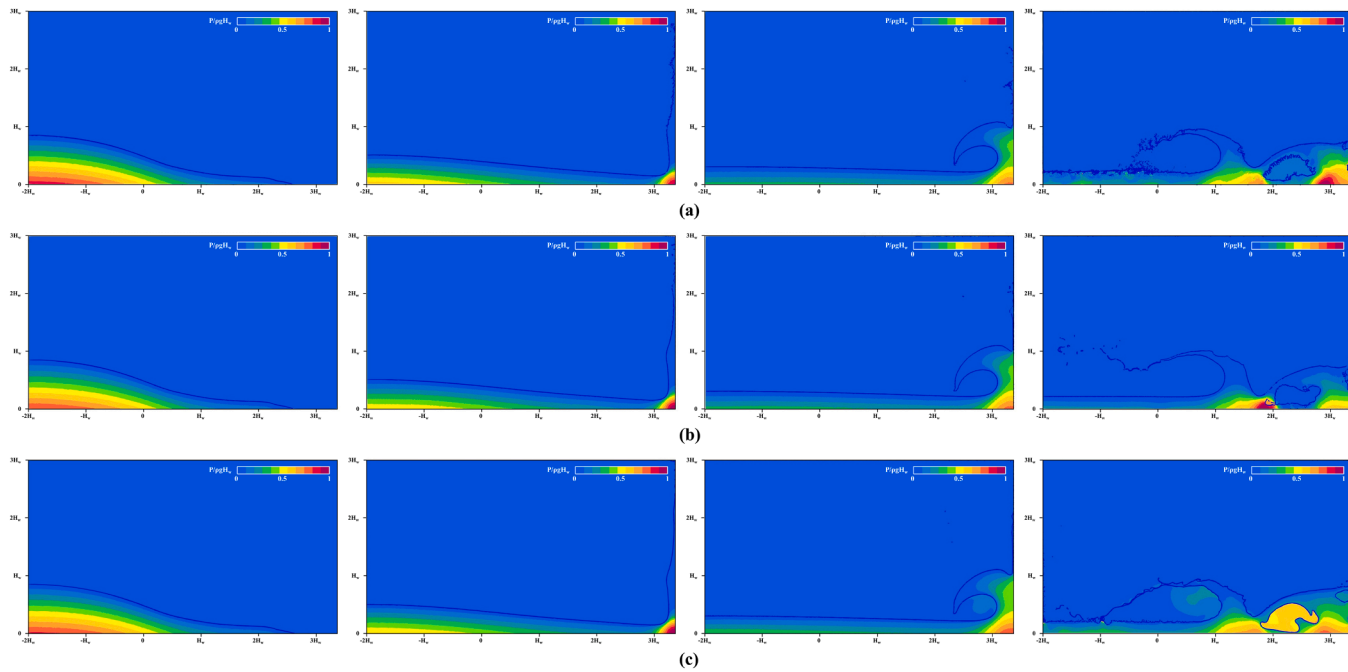


Fig. 12. Two-phase 2D dam break: snapshots over at different time instants $t/(g/H)^{0.5} = 0.0, 2.0, 4.0, 6.0, \text{ and } 8.0$ with $\Delta x = H_W/160$ obtained by (a) WCSPH, (b) EISPH and (c) EICSPH.

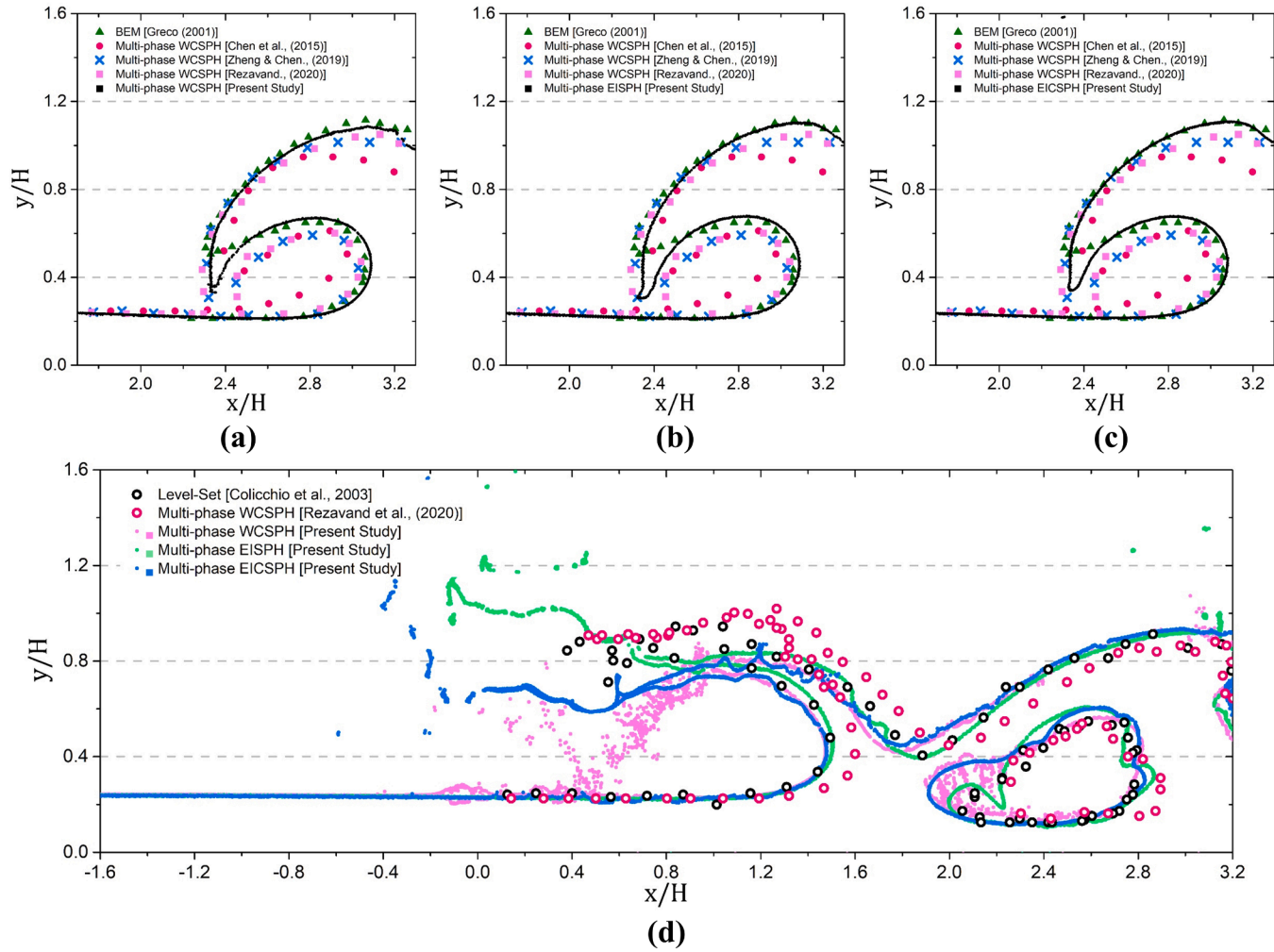


Fig. 13. Two-phase 2D dam break: numerical prediction of water-air interface at $t(g/H)^{0.5} = 6.0$ obtained by (a) WCSPH, (b) EISPH and (c) EICSPH and (d) at $t(g/H)^{0.5} = 7.14$ using three schemes.

particle shifting technique.

Fig. 13 compares the air-water interface profiles obtained by the three schemes with those obtained by a Boundary Element Method (BEM) [84] and Level-set method [85]. The interface profiles of the three schemes at $t(g/H)^{0.5} = 6.0$ are shown in Fig. 13a, b and c and are compared with other SPH results and the BEM result. The captured interface morphology is in good agreement with other numerical results, but some discrepancies exist with the BEM result, which is simulated using a strictly incompressible model [23]. The water-air interface obtained by the three schemes in this study shows a relatively better prediction compared to other SPH studies. The interface profiles of the three schemes at $t(g/H)^{0.5} = 7.14$ are compared with numerical results obtained by previous SPH study and Level-set method in Fig. 13d. All three schemes accurately captured the shape of the interface and the presence of air pockets. Three methods in the present study predict the location of the front side of jet, contrary to the slight time delay seen in previous result. Among the three methods, the strictly incompressible EISPH method showed the best agreement with both numerical and experimental results, followed by the EICSPH and WSPH methods, respectively. Furthermore, the EISPH and EICSPH methods demonstrated sharper interfaces compared to WSPH.

The pressure signals obtained at pressure probes P1 and P2 are compared with the experimental results [81] in Fig. 14. At the lower right corner where P1 is located, the first peak appears when the wave collides with the wall. The complex flow behavior is generated when the water rolls up along the wall and collided with water surface, resulting in a second smaller peak. The simulation results accurately predicted the pressure signals following the first peak for all three methods, but a slight delay was observed for the appearance of the second peak. The discrepancy is reasonable due to details of the initial condition in experiment and roughness of the walls [11]. The pressure measurement following the second peak showed significantly more noise in the WSPH method due to its acoustic noise compared to the EISPH and EICSPH methods, especially at low spatial resolution. However, it accurately followed the global trend and the three methods showed good convergence with the increase of spatial resolution. The pressure distribution measured at P2, located at a higher position, was also well predicted by all three methods.

The time evolution of water level measured at sensors H1 and H2 are compared with experimental data [82] in Fig. 15. The rise in water level from the moment the flow first reaches the sensors to the moment when the wave collides with the wall and returns is well predicted by all three methods. However, some differences between the numerical results and experimental results are shown after the moment when the flow becomes more complex. These discrepancies are also shown in previous studies and thought to be reasonable due to difference in initial conditions and boundary wall roughness. Nevertheless, the results obtained from the three different schemes show very similar trends.

The evolutions of mechanical energy over time are compared among the three methods in Fig. 16. Mechanical energy is normalized based on imbalance between initial potential energy and final potential energy [86]:

$$\Delta E = \frac{E_{pot} + E_{kin} - E_{pot}^0}{E_{pot}^0 - E_{pot}^{inf}}. \quad (45)$$

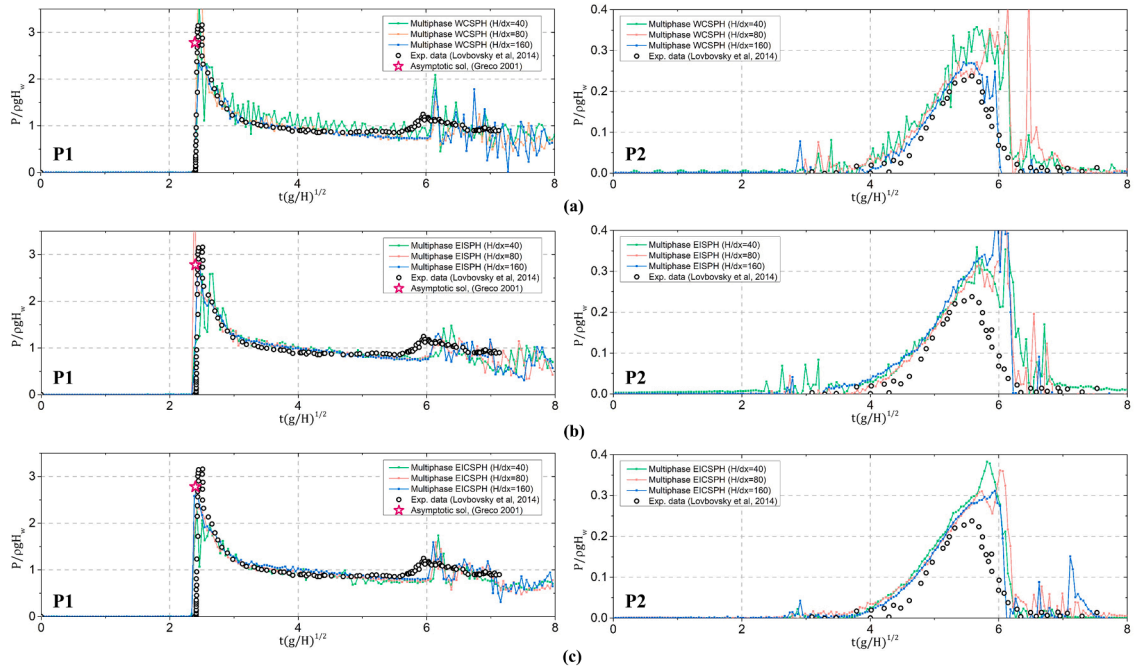


Fig. 14. Two-phase 2D dam break: the time history of the pressure signals at pressure probe P1, and P2 obtained by (a) WSPH, (b) EISPH and (c) EICSPH, in comparison with experimental data.

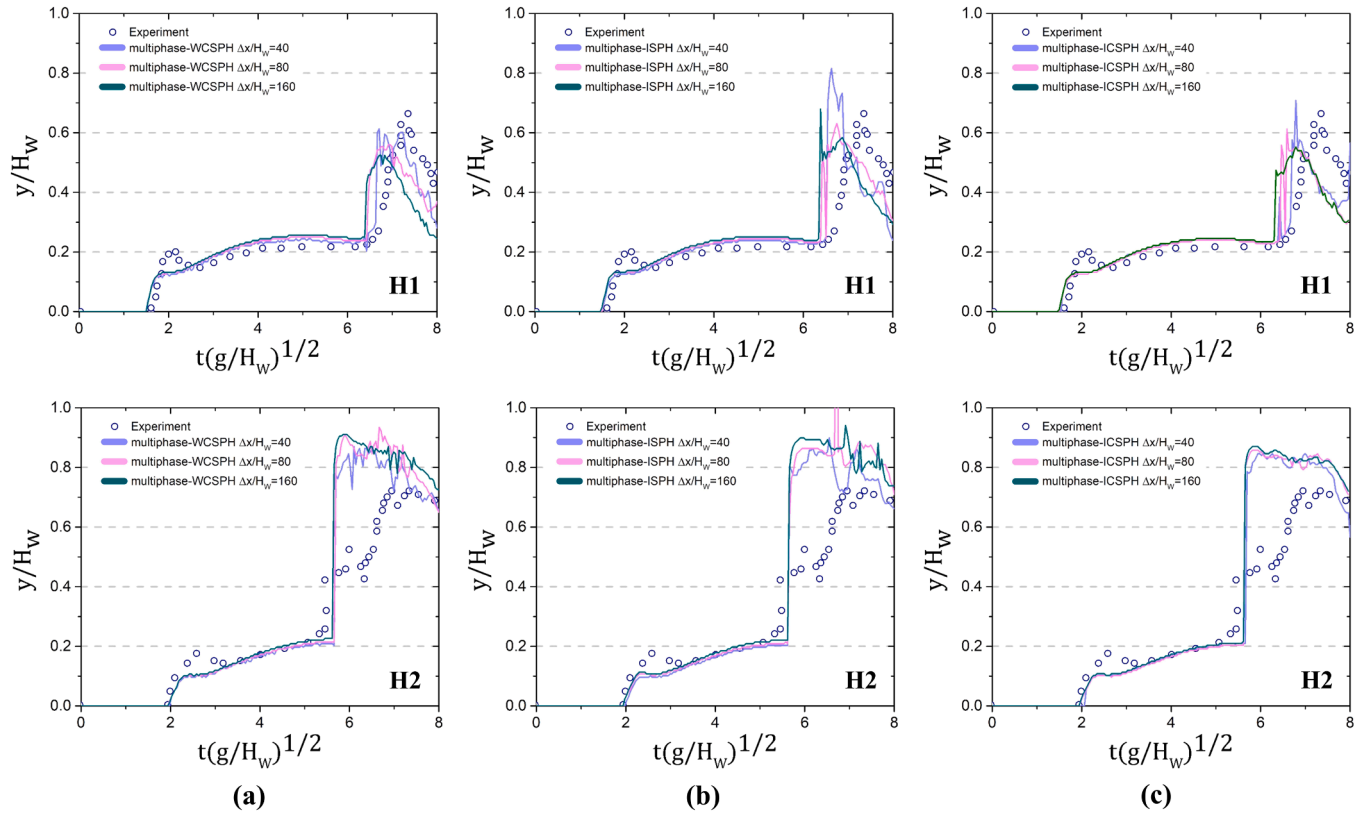


Fig. 15. Two-phase 2D dam break: the time history of water level at height measurement sensors H1, and H2 obtained by (a) WCSPH, (b) EISPH and (c) EICSPH, in comparison with experimental data.

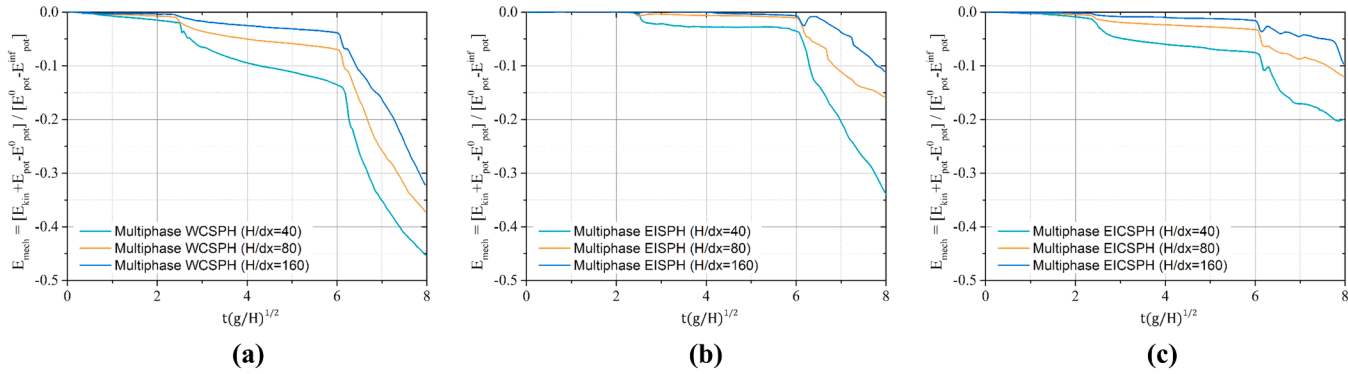


Fig. 16. Two-phase 2D dam break: the time history of numerical dissipation of mechanical energy obtained by (a) WCsPH, (b) EISPH and (c) EICSPH.

Here, E_{pot}^0 represents the initial mechanical energy, and E_{pot}^{inf} represents the mechanical energy that would occur when the flow reaches a hydrostatic state after coming to a standstill. Initially, as all particles are at rest, mechanical energy consists solely of potential energy. As the flow evolves, potential energy converts to kinetic energy, and some of it is dissipated through viscous dissipation. Overall, the energy gradually decays, with significant deviations observed when the flow initially collides at the right corner and generated water column collides with the free surface. Across all methods, higher resolution results in longer preservation of mechanical energy. As expected, WCSPH exhibits the highest numerical dissipation, while EICSPH demonstrates noticeably lower numerical dissipation compared to WCSPH and EISPH.

It may be of interest to explore the impact of air compressibility on dam break scenarios. For the case of EICSPH where water is modeled as incompressible and air as compressible, adjusting the stiffness of air is flexible. Previous study coupled incompressible SPH and compressible SPH utilized the actual speed of sound for air to compare pressure signals against experimental data, particularly in scenarios like object slamming [45]. In another study coupled incompressible SPH and compressible SPH, the dam break problem was solved with an air sound speed of 50 m/s which is smaller than actual sound speed and showed different morphologies compared to fully incompressible level set approach. It is observed that the air pocket appears quite smaller compared to the level set solution, and the angle of the water jet is also larger compared to the level set results [46]. Some studies emphasized the significance of using the real speed of sound for air in realistically simulating air-cushion effects using multi-phase WCSPH [11,66,87]. To investigate the effects of air speed of sound, we compared and visualized the results of the dam break scenario for three different air sound speeds: 34 m/s, 170 m/s, and 340 m/s. Fig. 17 illustrates the morphology of the free surface at time instants $t^* = 6.464$ and $t^* = 7.14$. As can be seen in Fig. 17a, when the speed of sound gets small, the jet forms at a larger angle. This may be attributed to the stronger air-cushion effect, resulting in the loss of mechanical energy. At time instant $t^* = 7.14$ in Fig. 17b, it becomes apparent that when the speed of sound is lower than the actual sound speed ($=340$ m/s), the jet fails to advance further. Aligning the speed of sound for air with the actual value yields results similar to the level set solution, which assumes both water and air are incompressible. Conversely, setting the speed of sound for air lower than the actual value reveals differences, particularly in the size of the air pocket. With increasing air compressibility, the air pocket undergoes more pronounced compression, indicating a strong air cushion effect. When considering a sound speed of air at 34 m/s, the morphology of the free surface closely resembles that observed in a previous study utilizing the ICSPH method with a sound speed of air as 50 m/s which was also deviated from level set solution [45]. The dissipation of kinetic energy can be observed in Fig. 18a. The decrease in mechanical energy over time is plotted for cases where the speed of sound is 34 m/s and 340 m/s. Notably, when the speed of sound is 34 m/s, a sharp decline in mechanical energy is observed immediately after the collision of the water column with the free surface at $t^* = 6$. Based on these findings, it can be emphasized that employing the actual sound speed is crucial for accurately capturing the real air cushion effect in practical scenarios. Despite variations in the shape of the free surface, the pressure evolution at the bottom right corner over time remains consistent across all three cases, showing good agreement with experimental results as in Fig. 18b.

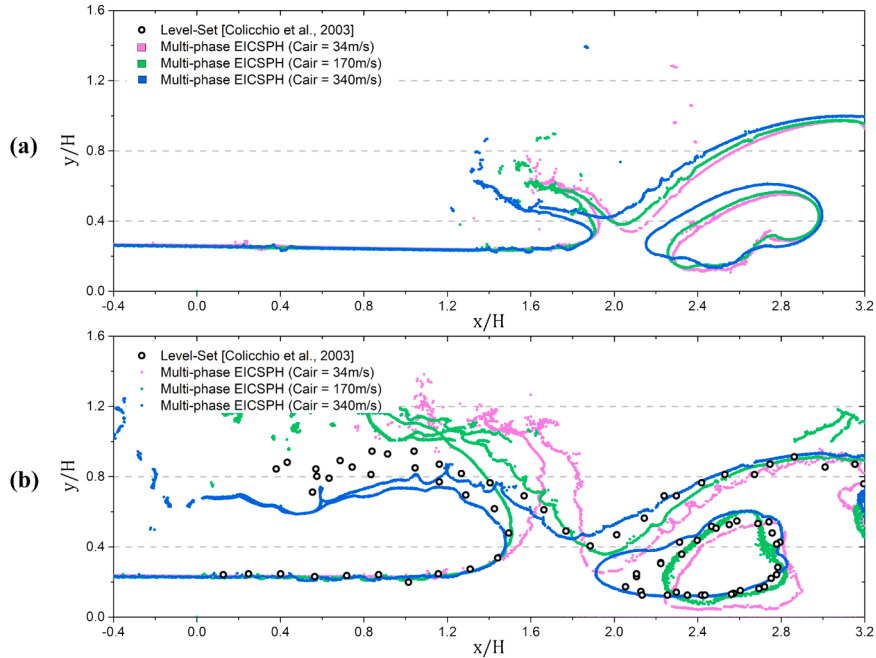


Fig. 17. Two-phase 2D dam break: numerical prediction of water-air interface at (a) $t(g/H)^{0.5} = 6.464$ and (b) at $t(g/H)^{0.5} = 7.14$ obtained by EICSPH scheme using three different air sound speeds.

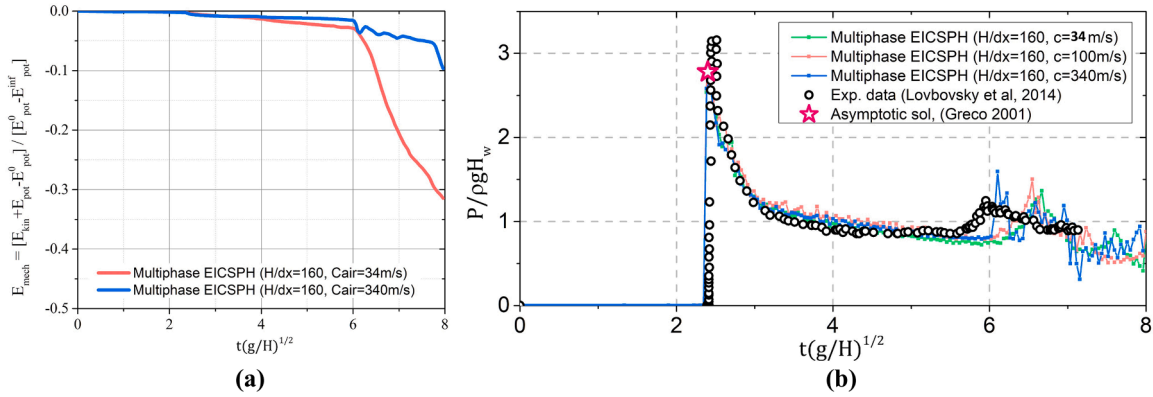


Fig. 18. Two-phase 2D dam break: (a) the time history of numerical dissipation of mechanical energy and (b) pressure signals at pressure probe P1 obtained by EICSPH scheme using different air sound speeds.

4.5. Computational efficiency

In this subsection, we discuss the computational time required to solve the transient problems of Rayleigh–Taylor instability, sloshing, and dam break. Fig. 19 illustrates the computational time for these three problems at three different resolutions using three methods. As expected, WCSPH and EICSPH are more time-consuming compared to EISPH in all cases. However, the relative computational efficiency between WCSPH and EICSPH was problem-dependent.

Firstly, when dealing with Rayleigh–Taylor instability, EICSPH handles half of the computational domain with WCSPH, and the other half with EISPH. However, the time difference between WCSPH and EISPH is not significant in this case, and the algorithm complexity increases during the coupling process, resulting in slightly longer computational time for EICSPH compared to WCSPH.

For sloshing and dam break, the computational performance of EISPH is significantly superior to that of WCSPH, which uses the actual sound speed for air. In the case of sloshing, since half of the computational domain consists of air particles, there may be an improvement in computational performance. However, due to the increased algorithm complexity, EICSPH shows slightly faster computational performance compared to WCSPH. The different characteristics in computational performance compared with Rayleigh–Taylor instability may be attributed to the significant time difference between EISPH and WCSPH, which offsets the increase in algorithm complexity. For the dam break problem where there are relatively more air particles, EICSPH shows similar computational performance to WCSPH.

5. Summary and conclusion

In this study, we proposed and compared three multi-phase SPH methods that have the ability to handle high-density differences in violent multi-phase flows. Our first method involves reformulating the conventional WCSPH approach as a quasi-Lagrangian framework based on the ALE context. Additionally, we extended the efficient and accurate EISPH method to handle multi-phase flows with numerical efficiency and high accuracy. Lastly, we combined these two approaches to develop the EICSPH method, which has a fully-explicit algorithm and can handle both compressible and incompressible phases. It is worth noting that EISPH and EICSPH are also written in a quasi-Lagrangian framework.

To validate the efficacy of the proposed methods, we conduct four benchmarks. The study initially addressed the Rayleigh–Taylor instability, where all three methods accurately tracked the intricate interface, demonstrating convergence even with small density differences (less than 2). Subsequently, the investigation extended to scenarios with larger density differences, examining static hydrostatic problems to verify stability. Further analysis included dynamic sloshing and dam break problems under large density difference. All three methods show good agreement in the pressure distribution of the multi-phase hydrostatic problem, with EICSPH demonstrating faster convergence compared to WCSPH and EISPH showing significantly faster convergence. Nevertheless, due to the fundamental discontinuity, the kinetic energy of the air phase of EISPH was observed to converge relatively high compared to WCSPH and EICSPH. Regarding the sloshing problem, all three methods accurately predicted the wave height induced by forced oscillations over time. Although minor pressure noise was initially observed in WCSPH, all three methods demonstrated converging pressure trends over time, closely aligning with experimental results. In the dam break problem, the results of the three methods were compared with experimental results and other incompressible numerical methods. All three methods successfully simulated sharp interfaces in complex flows without non-physical voids. The time-varying pressure and height were well predicted, and it was confirmed that the EISPH, which is entirely incompressible, better matched the results obtained from other incompressible numerical methods compared to WCSPH and EICSPH. In the case of EICSPH, dam break results were examined under varying sound speeds of air. Minimal differences were noted in the pressure evolution over time compared to both the actual sound speed and cases with smaller sound speed. However, notable discrepancies were observed in the progression of the free surface. As the sound speed decreased, the air cushion effect in the generated air pocket expanded, leading to increased energy dissipation and interface motion. Hence, accurately

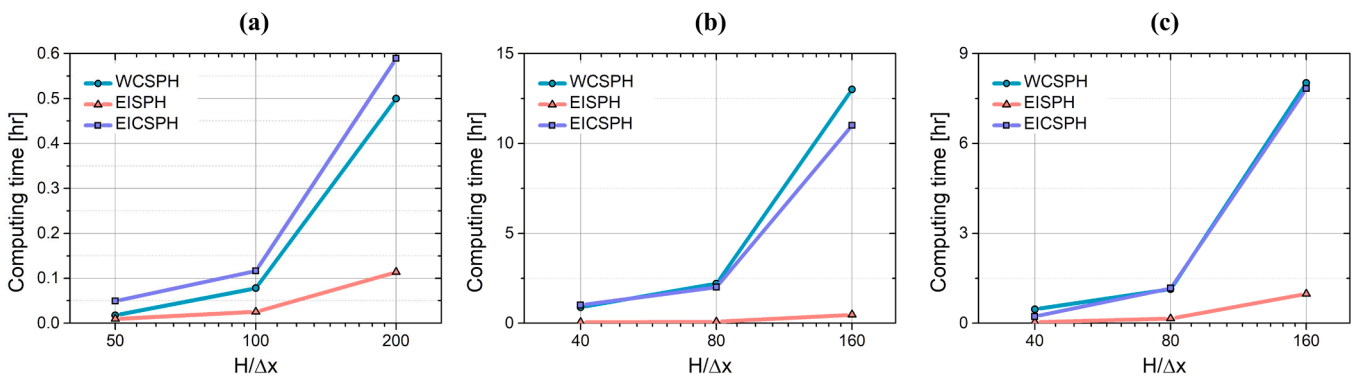


Fig. 19. Computing time of three transient benchmarks (a) Rayleigh–Taylor instability (b) Liquid Sloshing and (c) dam-break problem.

representing the real sound speed is crucial for effectively capturing the physical air cushion effect. Finally, a comparison of the computational times for the transient problems among the three methods was conducted. EISPH significantly outperformed the others in terms of computational efficiency. EICSPH, by combining the WCSPH with fast EISPH, could potentially offer time benefits, but it remained problem-dependent due to the increased algorithm complexity.

In conclusion, EISPH consistently demonstrated the most accurate results and computational efficiency. EICSPH followed, offering a decrease in acoustic noise present in WCSPH by coupling EISPH, but it was observed that the sound speed of air significantly influences the flow behavior. For violent incompressible multiphase flows such as those addressed in this study, we recommend to use the actual sound speed of air. Moreover, this study highlights the flexibility of EICSPH, allowing for one phase as incompressible while freely adjusting stiffness for the other, offering promising applicability and flexibility for future research.

CRedit authorship contribution statement

Hee Sang Yoo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Young Beom Jo:** Writing – review & editing, Software, Methodology. **Eung Soo Kim:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Eung Soo Kim reports financial support was provided by National Research Foundation of Korea.

Data availability

Data will be made available on request.

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